HUMAIRA ANZUM NEHA

Dhaka, Bangladesh | UIU Email | Personal Email
Google Scholar | LinkedIn | GitHub | Portfolio

ABOUT ME

Computer Science lecturer and researcher with experience in human-centered AI, computer vision, and accessibility, and strong interests in multimodal AI and natural language processing. My research focuses on designing and evaluating intelligent systems that better support diverse users, particularly those whose needs are often overlooked in conventional technology design.

EDUCATION

2026–Present **M.Sc. in Computer Science and Engineering** — **BRAC University**
CGPA: 4.00/4.00; 12 credits completed

2019–2024 **B.Sc. in Computer Science and Engineering** — **Khulna University of Engineering & Technology (KUET)**
CGPA: 3.87/4.00 (3rd Position)

RECENT RESEARCH PROJECTS

2026
2nd author

AI-Assisted Speech Editing for People Who Stutter
Manuscript under review.

- Conducted a qualitative HCI study with eight people who stutter in Bangladesh on video creation and AI-assisted speech editing, using an onboarding workshop and semi-structured interviews.
- Examined how video context, audience, and personal preferences shaped decisions to preserve or modify stuttering.
- Identified the need for AI support that reduces editing effort while preserving creator control and voice representation.

RESEARCH INTERESTS

Human–Centered AI, Mutlimodal AI, Human–Computer Interaction, Computer Vision, Natural Language Processing, Accessibility

SKILLS

Programming: Python, C/C++, Java, JavaScript, SQL

AI & ML: PyTorch, OpenCV, Hugging Face Transformers, Scikit-Learn

AI Tools: OpenAI Codex, Claude

WORK EXPERIENCE

2024–Present **Lecturer in Department of CSE**
United International University

- Courses Taught: Database Management Systems, Data Mining, Object Oriented Programming, Computer Architecture, Data Structures and Algorithms.

RESEARCH EXPERIENCE

2024
1st author

Automated Object Identification for the Visually Impaired
Published in the 2024 27th International Conference on Computer and Information Technology (ICCIT).

- Developed an edge-compatible assistive system for common-object detection, Bangladeshi currency and denomination recognition, and face recognition.
- Collected and annotated custom image datasets for model training and evaluation: object-detection dataset and Bangladeshi currency dataset.
- Technologies Used: Python, PyTorch, OpenCV, YOLOv5.

2024

An Assistive Device for the Visually Impaired for Enhanced Navigation and Reading

Published in the 2024 13th International Conference on Electrical and Computer Engineering (ICECE).

- Developed embedded spectacle glasses with real-time obstacle avoidance, fall detection, and textual information extraction modules.
- Technologies Used: Python, PyTorch, OpenCV, YOLOv5, Raspberry Pi, OCR, Accelerometer, and ToF Distance Sensor.

2023
1st author

Mobile Application to Collect Data and Measure Blood Component Levels in a Non-invasive Way

Published in the 2023 26th International Conference on Computer and Information Technology (ICCIT).

- Developed a mobile application for non-invasive measurement of blood components.
- The user places their fingertip on an NIR LED device, and a 15-second video is recorded. The video is analyzed to extract PPG signals, which are processed through a DNN model to predict glucose and hemoglobin levels.
- Technologies Used: Kotlin, Android Studio, Python, Photoplethysmography (PPG) Signal Analysis, and OpenCV. Repository

OPEN DATASETS AND RESEARCH RESOURCES

2024

Computer Vision Datasets

Assistive Object Detection Dataset (approximately 21,000 YOLO-annotated images across eight classes) and Bangladeshi Currency Classification Dataset.

ACADEMIC PROJECTS

Python, PyTorch, VLMs

Training-Free Long Video Understanding

Developed a training-free, query-guided asymmetric grid composition method for long video understanding with vision-language models (VLMs)

Python, Pygame

Abalone AI

Developed a playable Abalone board game in Pygame with an AI opponent powered by Minimax and alpha–beta pruning.

Bootstrap, JavaScript, ASP.NET

Smart&Safe

Developed a web app for buying and selling used smartphones with identity verification (NID + security information) and automated IMEI screening through an admin dashboard.

Java, Firebase, Android Studio

Together

Created a task and deadline management mobile application with collaboration and communication features for teams and coworkers.

ACHIEVEMENTS

3 times

Dean's List Award

Awarded for consistent academic excellence and strong semester-wise academic performance.

Khulna University of Engineering & Technology (KUET)

2021

Top-10 Finalist

Designed and prototyped a community-based application idea selected among the hackathon's top 10 entries.

UNDP Women's Digital Innovation Hackathon 2021

2018–2019 to 2021–2022

Top-3 CGPA (among 120 students)

Ranked in the top 3 by cumulative academic performance within the cohort.

Khulna University of Engineering & Technology (KUET)